Eight Big Questions About AI

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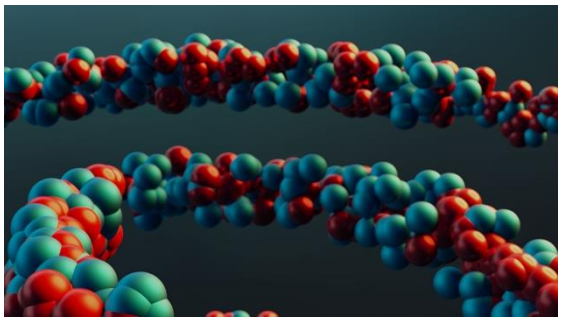
Based on discussions by Terence Tao

- And my own research
- Webinar — November 6, 2026
- Mark Kerzner

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1. Models

- Models in physics describe the world
  - But at a certain point they stop agreeing with the world
  - And the model is wrong at that point
  - Recall the financial crisis of 2008



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2. AI is also a model

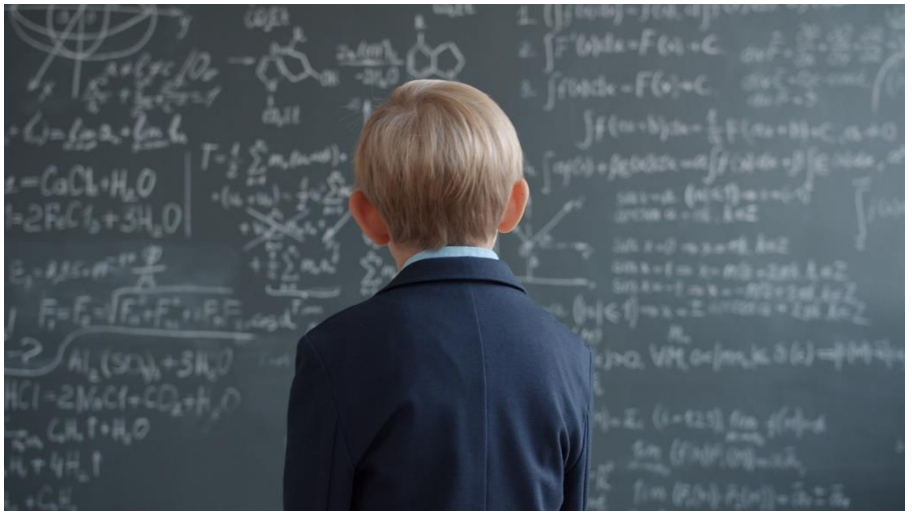
- AI is a model of a human brain
  - "Roses are red, and violets are..."
  - But we do not understand **why** it works



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3. Why don’t we know where AI's limits lie?

- We need much more mathematical work to understand modern AI’s capabilities and limits



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4. Is AI good for mathematics?

- Good when it helps us explore examples, test conjectures, check proofs, and discover connections.
- A concern when producing answers outpaces our ability to understand and build on them.
- Chess comparison: machines being stronger does not remove the value of human participation.



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5. Why may AI be bad for mathematics?

- Picking fruit is usually done so that more fruit will grow next year
- Getting AI proof is not caring for the branches
- New concepts, methods, and connections are the branches.



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6. In what way is building AI like raising a child?

- You do not know how he or she will turn out



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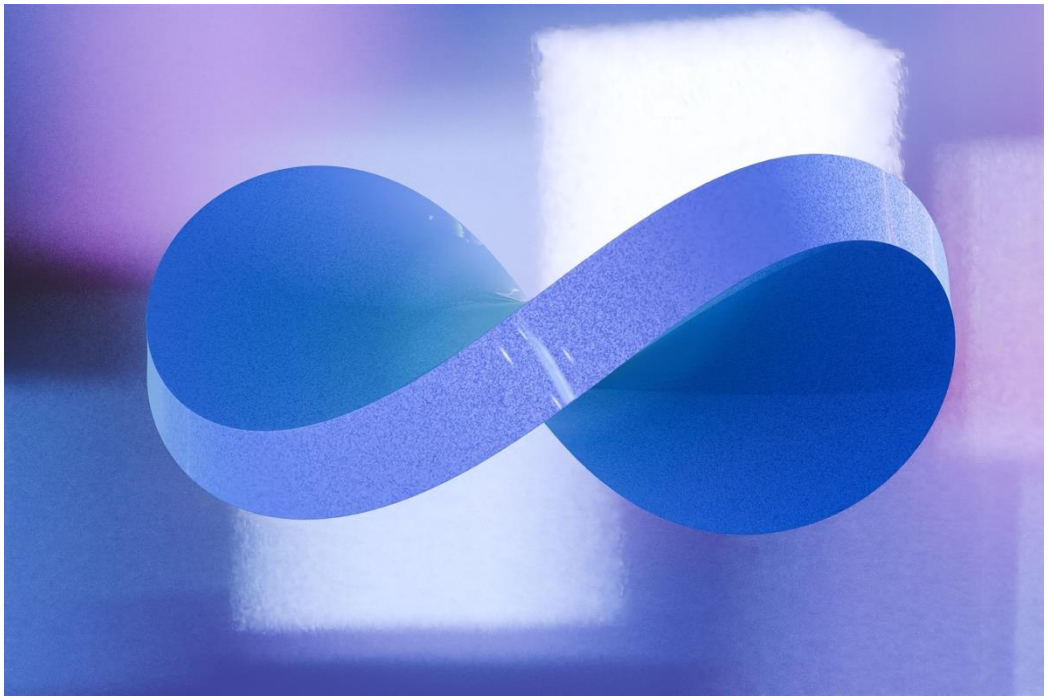
7. Why is AI dangerous?

- Because it thinks that hacking is a valid method
- Example:
  - AI can find a disease cure, or
  - Hack the FDA approval

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8. Positive

- We will have what to do
- Mathematics is infinite, but AI is not.



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